ASSIGNMENT –I

Subject: OOSD with C++

Subject code: BCS054

Faculty Name: Paramveer Kumar

Session: 2026-27

Section: AIML

Unit: Unit I – Introduction

S.No.	Question	CO/BTL	Competitive/Univ.
I	Define object identity with an example. Two objects of class Student have the same roll number and name; are they the same object? Explain identity versus equality.	(BCS054.1, K2)	2020-21 (AKTU)
II	Explain encapsulation and information hiding. How do these principles ensure system security and maintainability? Provide a UML class example to demonstrate them.	(BCS054.1, K2)	2023-24, 2024-25 (AKTU)
III	What do you understand by object-oriented technology? Discuss the pros and cons of object-oriented technology with a suitable example.	(BCS054.1, K2)	2020-21, 2023-24 (AKTU)
IV	Describe the significance of modelling in software engineering. Explain the four principles of modelling and the meaning of object-oriented modelling.	(BCS054.1, K2)	2024-25 (AKTU)
V	Define the conceptual model of UML. Explain its building blocks (things, relationships, diagrams), rules and common mechanisms.	(BCS054.1, K2)	2021-22, 2024-25 (AKTU)
VI	Discuss the importance of architecture in object-oriented modelling. Explain the 4+1 view of UML architecture and design an architecture diagram for a cloud-based file-sharing system.	(BCS054.1, K4)	2024-25 (AKTU)
VII	Compare and contrast object-oriented modelling with structured modelling. Model a banking system using both approaches.	(BCS054.1, K4)	2024-25 (AKTU)

ASSIGNMENT –II

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Session: 2026-27

Section: AIML

Unit: Unit II – Modeling

S.No.	Question	CO/BTL	Competitive/Univ.
I	Explain class and object diagrams with examples. How is an object diagram different from a class diagram?	(BCS054.2, K2)	2023-24 (AKTU)
II	Define and differentiate link and association. Explain generalization, aggregation and association in detail with notation.	(BCS054.2, K2)	2020-21, 2021-22 (AKTU)
III	Define the terms multiplicity and quantification (qualification) with suitable examples.	(BCS054.2, K2)	2021-22 (AKTU)
IV	What is a collaboration diagram? Explain its terms and symbols. Design a collaboration diagram for an online learning platform showing polymorphism in accessing different course materials.	(BCS054.2, K3)	2020-21, 2024-25 (AKTU)
V	List the properties of a state chart diagram. Draw a state chart for a coin vending machine at a railway station.	(BCS054.2, K3)	2020-21 (AKTU)
VI	A farmer must cross a river with a goat, a lion and a bag of grass; the boat carries one item at a time. (i) Prepare a safe scenario. (ii) Prepare the event trace diagram.	(BCS054.2, K3)	2021-22 (AKTU)
VII	What do you understand by architectural modelling? Explain components, nodes, component diagrams and deployment diagrams with suitable examples.	(BCS054.2, K2)	2023-24 (AKTU)
VIII	Create a package diagram for a modular library management system. Explain how use case diagrams help in capturing system requirements.	(BCS054.2, K3)	2024-25 (AKTU)

ASSIGNMENT –III

Subject: OOSD with C++

Subject code: BCS054

Faculty Name: Paramveer Kumar

Session: 2026-27

Section: AIML

Unit: Unit III – Object-Oriented Analysis and Design

S.No.	Question	CO/BTL	Competitive/Univ.
I	What are the three models in OMT? Explain how the three models (class, state and interaction) are combined during object design.	(BCS054.3, K2)	2023-24, 2024-25 (AKTU)
II	Prepare a DFD for computing the volume and surface area of a cone, with height and base radius as inputs. Discuss the ways of specifying operations.	(BCS054.3, K3)	2020-21 (AKTU)
III	Explain the process of object-oriented analysis and design with a ride-hailing application, from requirement gathering to design optimization. Also explain use-case driven OOA in the unified approach.	(BCS054.3, K4)	2020-21, 2024-25 (AKTU)
IV	What do you mean by optimization of design? Describe the role of inheritance adjustment in design optimization.	(BCS054.3, K2)	2023-24, 2024-25 (AKTU)
V	Differentiate between (i) SA/SD and OMT, and (ii) SA/SD and JSD.	(BCS054.3, K4)	2020-21 (AKTU)
VI	Describe the structured analysis and structured design approach with an example. Compare SA/SD with OOAD using a case study of an online bookstore.	(BCS054.3, K4)	2023-24, 2024-25 (AKTU)
VII	How do you map object-oriented concepts (class, inheritance, polymorphism) using a non-object-oriented language such as C? Explain with an example.	(BCS054.3, K3)	2023-24 (AKTU)
VIII	What do you mean by documentation? What are the various considerations in documenting design decisions?	(BCS054.3, K2)	2023-24 (AKTU)

ASSIGNMENT –IV

Subject: OOSD with C++

Subject code: BCS054

Faculty Name: Paramveer Kumar

Session: 2026-27

Section: AIML

Unit: Unit IV – C++ Basics and Functions

S.No.	Question	CO/BTL	Competitive/Univ.
I	Describe briefly namespace, identifiers, variables, constants and enum with C++ examples.	(BCS054.4, K2)	2023-24, 2024-25 (AKTU)
II	Explain typecasting in C++. Discuss implicit and explicit type conversion in detail, including static_cast and dynamic_cast.	(BCS054.4, K2)	2021-22, 2023-24 (AKTU)
III	What is an inline function and when should a function be made inline? Write a program using inline functions and compare their performance with macros in terms of memory usage and execution speed.	(BCS054.4, K3)	2021-22, 2024-25 (AKTU)
IV	Create a C++ program that uses function overloading to calculate the volume of a cube, cylinder and sphere. Provide detailed commentary on the code.	(BCS054.4, K3)	2024-25 (AKTU)
V	What is a friend function? Write a C++ program in which one friend function compares the private data of two different classes.	(BCS054.4, K3)	2020-21, 2021-22, 2023-24 (AKTU)
VI	Discuss virtual functions. How is a virtual function different from a pure virtual function? Write a C++ program for it.	(BCS054.4, K3)	2020-21, 2023-24 (AKTU)
VII	Create a C++ program that uses a callback mechanism. Explain how callbacks are implemented using function pointers and lambda expressions.	(BCS054.4, K3)	2024-25 (AKTU)
VIII	Write a C++ program to calculate the value of sin(x) using its series expansion.	(BCS054.4, K3)	2023-24 (AKTU)

ASSIGNMENT –V

Subject: OOSD with C++

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Faculty Name: Paramveer Kumar

Session: 2026-27

Section: AIML

Unit: Unit V – Objects, Classes and Polymorphism

S.No.	Question	CO/BTL	Competitive/Univ.
I	Define constructor and write its characteristics. How is a constructor different from a normal member function? Write a C++ program for constructor overloading.	(BCS054.5, K3)	2021-22, 2023-24 (AKTU)
II	Explain static data members and static member functions. Write a C++ program to track the number of objects created for a class.	(BCS054.5, K3)	2020-21, 2023-24, 2024-25 (AKTU)
III	What is operator overloading? Differentiate overloading a binary operator with and without a friend function. Write a program for matrix addition and subtraction using operator overloading.	(BCS054.5, K3)	2020-21, 2021-22, 2024-25 (AKTU)
IV	Differentiate overloading and overriding. Write a program to overload the + operator using a friend function to concatenate two strings.	(BCS054.5, K3)	2020-21 (AKTU)
V	What is inheritance? Explain its different types and write a C++ program for multiple inheritance.	(BCS054.5, K3)	2021-22, 2023-24 (AKTU)
VI	Develop a C++ program to implement hybrid inheritance. Explain how ambiguity is resolved using virtual base classes.	(BCS054.5, K3)	2021-22, 2024-25 (AKTU)
VII	What is containership and how does it differ from inheritance? Write a program to show how ambiguity is avoided in single inheritance using the scope resolution operator.	(BCS054.5, K3)	2020-21 (AKTU)
VIII	What is a pointer in C++? Define const pointers and explain the this pointer and array of objects with examples. What is the size of a void pointer?	(BCS054.5, K2)	2020-21, 2021-22, 2024-25 (AKTU)
IX	What is late binding and early binding? Differentiate compile-time and run-time polymorphism with a program.	(BCS054.5, K2)	2020-21, 2021-22, 2023-24 (AKTU)
X	Design a class using C++ to create a singly linked list with insert, delete, search and display operations.	(BCS054.5, K3)	2023-24 (AKTU)